

Algebra II Functions and Transformations Cheatsheet

Module 2

Parent Functions

Family	Parent	Key Feature
linear	$f(x) = x$	constant rate
quadratic	$f(x) = x^2$	vertex
cubic	$f(x) = x^3$	origin symmetry
absolute value	$f(x) = x $	corner vertex
square root	$f(x) = \sqrt{x}$	starts at endpoint
reciprocal	$f(x) = \frac{1}{x}$	asymptotes
exponential	$f(x) = b^x$	horizontal asymptote
logarithmic	$f(x) = \log_b x$	vertical asymptote

Transformation Form

$$g(x) = a f(b(x - h)) + k$$

h = horizontal shift, k = vertical shift

$|a|$ = vertical stretch/compression, $a < 0 \Rightarrow$ reflect over x -axis

horizontal scale factor = $\frac{1}{|b|}$, $b < 0 \Rightarrow$ reflect over y -axis

$|b| > 1 \Rightarrow$ horizontal compression, $0 < |b| < 1 \Rightarrow$ horizontal stretch

Domain and Range

domain = allowed inputs

range = possible outputs

Check denominators, even roots, logarithms, graph endpoints, and asymptotes.

Composition

$$(f \circ g)(x) = f(g(x))$$

The domain of $f \circ g$ must satisfy:

$$x \in \text{domain of } g, \quad g(x) \in \text{domain of } f$$

Inverses

$$f(f^{-1}(x)) = x, \quad f^{-1}(f(x)) = x$$

To find an inverse:

1. Write $y = f(x)$.
2. Switch x and y .
3. Solve for y .
4. State domain and range restrictions.

Horizontal line test: a function has an inverse function if every horizontal line hits the graph at most once.

Piecewise Functions

$$f(x) = \begin{cases} 2x + 1, & x < 0 \\ x^2, & x \geq 0 \end{cases}$$

Use the rule whose condition contains the input.

Quick Checks

- Transform inside the input affects x -values.
- Transform outside the function affects y -values.
- Inverse graphs reflect across $y = x$.
- Piecewise boundaries control which rule applies.